SOFTWARE REQUIREMENTS SPECIFICATION (Addendum)

Project: Oasis Resort Katpana - Resort Booking & Management System

Version: (Updated with CMS, Experiential Add-ons, and Localized Payments)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the dynamic booking website and Content Management System (CMS) for Oasis Resort, located in Katpana, Skardu, Gilgit-Baltistan. The system reduces reliance on third-party OTA platforms (like Booking.com) by allowing direct guest bookings and localized payment processing, while giving the resort owner complete control over website content.

1.2 Project Scope

The system is a full-stack web application built on React.js, Node.js/Express.js, and a MySQL relational database. It supports three user roles:

* **Guest:** Can view resort content, search available rooms, add experiential services (dining, spa, transport), pay securely via local methods, and leave verified reviews.

* **Resort Admin:** Has complete CMS control to update front-page content (testimonials, team members), manage room inventory and seasonal pricing, moderate reviews, and track all bookings via a central dashboard.

* **Reception Staff:** Can view upcoming check-ins/check-outs and update booking statuses via a mobile-optimized interface.

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2. Information Architecture & Sitemap

The website follows a hierarchical layout structured into public content pages, experiential modules, and a multi-step booking checkout engine.

* **1. Home:** Hero banners, quick search widget, resort overview, and verified testimonials.

* **2. About Us:** Resort history, vision, team founders, and awards.

* **3. Accommodation:** Room categories (Villas, Suites, Cottages), 360° virtual tours, and detailed amenities.

* **4. Dining & Cuisine:** Restaurant overviews, downloadable menus, and private dining bookings.

* **5. Activities & Experiences:** Jet skiing, desert safaris, spa wellness, and Balti cultural experiences.

* **6. Special Offers:** Seasonal promotions, honeymoon packages, and group discounts.

* **7. Booking Funnel:** Date selection, dynamic pricing, add-on selection, and secure checkout.

* **8. Policies & FAQs:** Cancellation terms, resort rules, and automated booking assistant.

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3. Functional Requirements

3.1 Website Content Management (Admin Dashboard)

ID	Requirement	Priority
FR-CMS.1	The Admin shall be able to create, edit, and delete customer Testimonials and Team Member profiles directly from the dashboard without code changes.	High
FR-CMS.2	The Admin shall be able to create and manage blog posts, resort events, and update the FAQs and policy documents.	Medium
FR-CMS.3	The Admin shall be able to manage promotional codes (percentage or flat discount), setting usage limits and valid date ranges.	High

3.2 Room Inventory & Pricing Management

ID	Requirement	Priority
FR-ROOM.1	The Admin shall be able to add, edit, and deactivate room listings (e.g., Luxury Villas, Cottages) along with their base pricing and high-resolution images.	

| High |
| **FR-ROOM.2** | The system shall support Seasonal Dynamic Pricing, allowing the Admin to define specific date ranges (e.g., Peak Summer) that override the standard base price. | High |
| **FR-ROOM.3** | The room detail page shall display up to three related-room recommendations to prevent user cognitive overload.

| Medium |

3.3 Booking Engine & Checkout

ID	Requirement	Priority
FR-BOOK.1	The system shall require the server to re-verify room availability using MySQL row-level locks before confirming any booking to prevent double-booking.

| High |
| **FR-BOOK.2** | Guests shall be able to select optional add-ons (Private Dining, Spa Sessions, Airport Transfers) during the checkout process.

| Medium |
| **FR-BOOK.3** | The checkout shall present all total prices, fees, and applied promo discounts transparently before payment.

| High |
| **FR-BOOK.4** | The system shall support multiple payment methods: Direct JazzCash/Easypaisa API integration, Bank Transfer (with receipt upload field), and Pay at Hotel. | High |

3.4 Reviews & Customer Trust

ID	Requirement	Priority
FR-REV.1	The system shall strictly limit review submissions (1-5 stars and comment) to guests who have a booking marked as "Completed".

| High |
| **FR-REV.2** | The system shall provide an FAQ assistant widget. If a guest asks a question not in the database, it must fallback to a "contact front desk" message rather than hallucinating an answer.

| High |

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4. System Architecture (MySQL Database Design)

The system relies on a normalized MySQL database to handle complex relationships between bookings, seasonal rates, and add-on services.

1. Rooms & Pricing Subsystem

* `rooms` (id, title, category, base_price, is_active)
* `room_seasonal_prices` (id, room_id, season_name, start_date, end_date, price_per_night)
* `offers` (id, code, discount_type, discount_value, valid_from, valid_until, usage_limit)

2. CMS Subsystem

* `team_members` (id, name, role, bio, image_url)
* `testimonials` (id, guest_name, rating, comment, is_approved)
* `add_on_services` (id, category, title, price, is_available)

3. Booking & Transaction Subsystem

* `bookings` (id, room_id, guest_name, check_in, check_out, base_room_total, add_ons_total, discount_amount, final_total, payment_method, payment_status, booking_status)
* `booking_add_ons` (id, booking_id, add_on_id, quantity, unit_price)

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5. Non-Functional Requirements

* **NFR-1 (Performance):** Core pages must load in under 3 seconds on a 3G-equivalent mobile connection to accommodate Gilgit-Baltistan's variable internet connectivity.

* **NFR-2 (Resilience):** The system shall retain in-progress checkout data locally so a guest does not lose their booking progress if the network drops temporarily.

* **NFR-3 (Security):** All API traffic must use HTTPS, passwords must be hashed via bcrypt, and payment webhook callbacks must be signature-verified server-side before updating a booking to "Paid".

* **NFR-4 (Usability):** The application shall follow a mobile-first responsive design, ensuring it is fully operational on screens as small as 320px wide.

1. Architectural Overview & Sitemap Hierarchy
The system follows a hierarchical layout structured into public content pages, experiential modules, and a dedicated multi-step booking checkout engine.

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├── 1. HOME
│   ├── Hero Section (Dynamic Banners / Imagery)
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- Quick Booking Form (Search widget: Dates, Guests, Category)
- Overview of Resort (Short description & highlights)
- Testimonials & Reviews (Verified guest feedback)

- 2. ABOUT US
  - Resort History
  - Vision & Mission
  - Team & Founders
  - Awards & Recognition

- 3. ACCOMMODATION
  - Room Categories (Luxury Villas, Suites, Cottages)
  - Room Details (Amenities, pricing, capacity, specs)
  - Virtual Tour (360° views / media gallery)
  - Booking CTA (Direct redirect to booking engine)

- 4. DINING & CUISINE
  - Restaurants & Bars
  - Menu Previews (Downloadable/viewable menus & prices)
  - Special Dining Experiences (Private Dining, Scenic/Riverside dining)
  - Food & Beverage Gallery

- 5. ACTIVITIES & EXPERIENCES
  - Water / Desert Sports & Excursions (Jet Ski, Boating, 4x4 Desert Safari)
  - Spa & Wellness
  - Adventure & Excursions (Trekking, Sightseeing)
  - Cultural Experiences (Local Balti heritage, music, crafts)
  - Resort Events (Bonfires, stargazing, gatherings)

- 6. MEDIA GALLERY
  - Categorized Media (Landscape, Rooms, Dining, Guest Highlights)
  - High-definition optimized asset viewer

- 7. SPECIAL OFFERS & PROMOTIONS
  - Seasonal Promotions (Peak vs. Off-peak packages)
  - Honeymoon & Wedding Packages
  - Corporate & Group Discounts

- 8. WEDDINGS & EVENTS
  - Destination Weddings & Packages
  - Event Planning Services
  - Meeting & Conference Spaces

- 9. BLOG & TRAVEL CONTENT
  - Travel Guides (Skardu itineraries, packing guides, weather updates)
  - Guest Experiences & Travel Stories

- 10. CONTACT US
  - Interactive Contact Form
  - Social Media Links & Direct Channels

- Location Pin & Quick Inquiries

- 11. BOOKING & CHECKOUT FUNNEL *(labeled as reservation flow in sitemap)*
  - Room Selection & Dynamic Pricing
  - Experience Add-Ons (Dining reservations, Spa packages, Tour activities)
  - Checkout & Payment Confirmation (JazzCash, Easypaisa, Bank Transfer, Pay at Hotel)

- 12. POLICIES & FAQS
  - Cancellation & Refund Policy
  - Resort Rules & Check-in Guidelines
  - Frequently Asked Questions